

210B Homework 0

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Problem 1. Let G be a group, $N \trianglelefteq G$, and assume that the quotient group G/N is cyclic. If $N \subseteq Z(G)$, prove that G is abelian. Here $Z(G)$ denotes the center of G .

Proof. Suppose $N \subseteq Z(G)$. Since $N \leq G$, we know that the left-cosets of N in G partition G . That is,

$$G = \bigcup_{g \in G} gN. \quad (*)$$

Since G/N is cyclic, it follows that there exists an element $z \in G$ such that $\langle zN \rangle = G$. Then $(*)$ can be written in the following way:

$$G = \bigcup_{n \in \mathbb{N}} (zN)^n = \bigcup_{n \in \mathbb{N}} z^n N.$$

Now, let $x, y \in G$. Then for some $i, j \in \mathbb{N}$, $x \in z^i N$ and $y \in z^j N$. Then for some n_1 and n_2 in $\mathbb{N}$, it follows that

$$x = z^i n_1 \text{ and } y = z^j n_2.$$

Since $N \leq Z(G)$, we have $n_1, n_2 \in Z(G)$; that is, z_1 and z_2 commute with every element of G . Namely, z_1 and z_2 commute with every element of G . Namely, z_1 and z_2 commute with z^i and z^j . Hence, after some rearrangements

$$xy = (z^i n_1)(z^j n_2) = (z^j n_2)(z^i n_1) = yx.$$

Thus, we get that $xy = yx$ and so G is abelian. ■

Problem 2. Let G be a group and let H be a subgroup of G . Let $N_G(H)$, $C_G(H)$ and $\text{Aut}(H)$ be the normalizer of H in G , the centralizer of H in G , and the group of automorphisms of H , respectively.

(a) For $g \in N_G(H)$, prove that $\varphi_g(x) = gxg^{-1}$ is an automorphism of H .

Proof. Our goal is to show that φ_g is a

(i) homomorphism and a

(ii) bijection

(i) Let $x_1, x_2 \in H$ be arbitrary. Then we have

$$\begin{aligned} \varphi_g(h_1 h_2) &= g(h_1 h_2)g^{-1} \\ &= (gh_1 g^{-1})(gh_2 g^{-1}) \\ &= \varphi_g(h_1) \varphi_g(h_2). \end{aligned}$$

Hence, we conclude that φ_g is a homomorphism.

(ii) Notice that $\varphi_g(H) = gHg^{-1}$. Since $g \in N_G(H)$, it follows that $gHg^{-1} = H$ and so $\varphi_g(H) = H$. Thus, we get that φ_g is a bijection from H to H . ■

(b) Prove that $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$.

Proof. Note that the automorphism proven in (a) represents the action of G on H by conjugation; that is, $g \cdot h = ghg^{-1}$ for all $g \in G$ and $h \in H$. It is straightforward to prove that the above is indeed an action on H . Now, consider the permutation representation

$$\psi : G \rightarrow S_H \quad g \mapsto \varphi_g. \quad (*)$$

Notice that since φ_g is a homomorphism, it also follows that ψ is a homomorphism. From the First Isomorphism Theorem, it follows that

$$G/\ker(\psi) \cong \psi(G) \leq \text{Aut}(H). \quad (**)$$

Furthermore, we have

$$\begin{aligned} \ker(\psi) &= \{g \in G : \psi(g)(h) = h \quad \forall h \in H\} \\ &= \{g \in G : \varphi_g(h) = h \quad \forall h \in H\} \\ &= \{g \in G : ghg^{-1} = h \quad \forall h \in H\} \\ &= C_G(H). \end{aligned}$$

Notice further that since φ_g is an automorphism from (a), it follows that for all $gHg^{-1} = H$ which means that H must be a normal subgroup of G . Letting $N_G(H) = G$ (since $gHg^{-1} = H$), we can rewrite (**) in the following way:

$$N_G(H) \setminus C_G(H) \cong \psi(G) \leq \text{Aut}(H)$$

and we are done. ■

Problem 3. Let p and q be distinct odd primes with $p < q$. Suppose p does not divide $q - 1$.

(a) Prove that a group of order p^2q must be abelian.

Proof. Let $|G| = p^2q$. Using Sylow's Theorem, we will analyze the number of sylow- p and sylow- q subgroups denoted by n_q and n_p . Let us start with n_p . Indeed, we have $n_p \equiv 1 \pmod{p}$ and $n_p \mid q$. That is, we have $n_p \in \{1, q\}$. If $n_p = q$, then $q \equiv 1 \pmod{p}$. This tells us that $q - 1 \equiv 0 \pmod{p}$ and so p divides $q - 1$ which is a contradiction. Thus, it follows that $n_p = 1$ and so we conclude that P is the unique normal subgroup of order p^2 .

Now, consider n_q . Then we have $n_q \equiv 1 \pmod{q}$ and $n_q \mid p^2$. This tells us that $n_q \in \{1, p, p^2\}$. If $n_q = p$, then we get

$$p \equiv 1 \pmod{q} \implies p - 1 \equiv 0 \pmod{q}$$

which tells us that $q \mid p - 1$ and so $q < p - 1 < p$ which is a contradiction. If $n_q = p^2$, then we have

$$\begin{aligned} p^2 &\equiv 1 \pmod{q} \implies p^2 - 1 \equiv 0 \pmod{q} \\ &\implies (p - 1)(p + 1) \equiv 0 \pmod{q}. \end{aligned}$$

This tells us that $q \mid p - 1$ and $q \mid p + 1$. But note that since p and q are odd primes such that $q > p + 1$, it follows that $q \nmid p + 1$. Furthermore, if $q \mid p - 1$, then we have $q < p$ which tells us that $q \nmid p - 1$. Since $q \nmid p + 1$ and $q \nmid p - 1$, we have a contradiction. So, it follows that $n_q = 1$ and so Q must be the unique Sylow- q subgroup of order q .

Because P and Q are normal subgroups of G , we have that $PQ \leq G$ and $P \cap Q = 1_G$ since p and q are relatively prime. This subsequently tells us that

$$|PQ| = \frac{|P||Q|}{|P \cap Q|} = \frac{|P||Q|}{1} = |P||Q| = p^2q = |G|$$

and $PQ \leq G$. Thus, we get that $G = PQ$. Since $P \cap Q = 1_G$ and $G = PQ$, it follows that for every $a \in P$ and $b \in Q$, $ab = ba$. Now, we need to show that G is abelian. To this end, let $x, y \in G$ be arbitrary. Then for some $x_1, x_2 \in P$ and $y_1, y_2 \in Q$, we have $x = x_1y_1$ and

$y = x_2y_2$. Then it follows that (after some rearrangement)

$$xy = (x_1y_1)(x_2y_2) = (x_2y_2)(x_1y_1) = yx.$$

Thus, we conclude that $xy = yx$ and so we have that G is abelian. ■

- (b) Give an example of a group order p^2q for each possible isomorphism class.

Solution. Let $p = 5$ and $q = 7$. Note that $5 \nmid 7 - 1 = 6$ and $p < q$. From (a), we can see that $G = PQ \cong P \times Q \cong \mathbb{Z}_{p^2} \times \mathbb{Z}_q$. Now we have the following isomorphism classes

$$\begin{aligned} G &\cong \mathbb{Z}_{5^2} \times \mathbb{Z}_7 \\ &\cong \mathbb{Z}_5 \times \mathbb{Z}_5 \times \mathbb{Z}_7. \end{aligned}$$
■

Problem 4. (a) Let G be a group and g be an element in G with order n . If k is a non-zero integer, prove that the order of g^k is $\frac{n}{\gcd(n, k)}$.

Proof. Denote $d = \gcd(n, k)$ and notice that there exists $q_1, q_2 \in \mathbb{Z}^+$ such that

$$n = dq_1 \text{ and } n = dq_2. \quad (*)$$

Furthermore, denote $|g^k| = a$. To show the above result, it suffices to show that

$$a \mid \frac{n}{d} \text{ and } \frac{n}{d} \mid a. \quad (\dagger)$$

We will first show that $a \mid \frac{n}{d}$. Notice from (*) that

$$(g^k)^{\frac{n}{d}} = g^{\frac{nk}{d}} = g^{\frac{n(dq_2)}{d}} = g^{nq_2} = (g^n)^{q_2} = 1_G.$$

Thus, it follows that $a \mid \frac{n}{d}$.

Now, we will show that $\frac{n}{d} \mid a$. Here denote $\ell = \frac{n}{d}$. We proceed by using the division algorithm to write

$$a = \ell q + r \text{ for some } q \in \mathbb{Z}^+ \text{ and } 0 \leq r < a.$$

Then we can see that (since the order of g is n)

$$\begin{aligned} (g^k)^a &= (g^k)^\ell = (g^k)^{\ell q} \cdot (g^k)^r \\ &= (g^k)^{\frac{n}{d}q} \cdot (g^k)^r \\ &= g^{nqq_2} \cdot (g^k)^r \\ &= (g^n)^{qq_2} \cdot (g^k)^r \\ &= 1 \cdot (g^k)^r. \end{aligned}$$

Thus, we have

$$(g^k)^{a-r} = 1.$$

Notice that we need $r = 0$ otherwise $r \neq 0$ would contradict the minimality of a (the order of g^k). Thus, we conclude that $\frac{n}{d} \mid a$. Hence, we have

$$|g^k| = \frac{n}{\gcd(n, k)}.$$
■

- (b) Assume G is a cyclic group of order 60. Let g denote a generator of G . Find all integers $k \in \{1, 2, \dots, 60\}$ such that g^k has order 6.

Proof. Using part (a), we get that

$$|g^k| = \frac{60}{\gcd(k, 60)} = 6.$$

Using the above, we obtain

$$60 = \gcd(k, 60) \iff 10 = \gcd(k, 60) \text{ where } k \in \{1, 2, \dots, 60\}.$$

This tells us that

$$10 = \gcd(10m, 60) \implies 10 = 10\gcd(m, 6) \implies 1 = \gcd(m, 6) \text{ where } 1 \leq m \leq 6.$$

The only choices of m for which the above holds is when either $m = 1$ or $m = 5$; that is, either $k = 10$ or $k = 50$. ■

Problem 5. Let G be a group of order $345 = 3 \cdot 5 \cdot 23$.

(a) Prove that G has a cyclic subgroup of order $3 \cdot 23 = 69$ and a cyclic subgroup of order $5 \cdot 23 = 115$.

Proof. We will start by show that there exists a cyclic subgroup of order $3 \cdot 23 = 69$. Consider the number of sylow 3-subgroups of G denoted by n_3 . By Sylow's Theorem, it follows that $n_3 \equiv 1 \pmod{3}$ and $n_3 \mid 5 \cdot 23$. This tells us that $n_3 = 5$, $n_3 = 23$, or $n_3 = 1$. If $n_3 = 5$, then

$$5 \equiv 2 \pmod{3} \not\equiv 1 \pmod{3} \implies n_3 \neq 5.$$

Likewise, if $n_3 = 23$, then we find that

$$23 \equiv 2 \pmod{3} \not\equiv 1 \pmod{3} \implies n_3 \neq 23.$$

Thus, the only option we have is for $n_3 = 1$. This tells us that P_3 is the unique Sylow 3-subgroup of order 3.

Next, let us look at the number of Sylow 23-subgroups n_{23} . Similarly, we will prove that $n_{23} = 1$. Using Sylow's Theorem, it follows that $n_{23} \equiv 1 \pmod{23}$ and $n_{23} \mid 3 \cdot 5$. Thus, $n_{23} \in \{1, 3, 5\}$. If $n_{23} = 3$, then

$$3 \equiv 3 \pmod{23} \not\equiv 1 \pmod{23} \implies n_{23} \neq 3.$$

Likewise, if $n_{23} = 5$, then

$$5 \equiv 5 \pmod{23} \not\equiv 1 \pmod{23}.$$

This tells us that $n_{23} = 1$ and so P_{23} is the unique subgroup of order 23 which is normal in G . Since P_3 and P_{23} are normal subgroups of G , it follows that $P_3 P_{23} \leq G$ and that $P_3 \cap P_{23} = 1_G$. Next, notice that

$$|P_{23}P_3| = \frac{|P_{23}||P_3|}{|P_{23} \cap P_3|} = \frac{|P_{23}||P_3|}{1_G} = |P_{23}||P_3| = 23 \cdot 3$$

and that (since 23 and 3 are relatively prime)

$$P_{23}P_3 \cong P_{23} \times P_3 \cong \mathbb{Z}_{23} \times \mathbb{Z}_3 \cong \mathbb{Z}_{23 \cdot 3}.$$

Notice that $\mathbb{Z}_{23 \cdot 3}$ is a cyclic subgroup which tells us that $P_{23}P_3$ also has to be a cyclic subgroup of the same order.

Now, let us prove that G has a cyclic subgroup of order $5 \cdot 23$. Let us look at the number of Sylow 5-subgroups denoted by n_5 . Using Sylow's Theorem, it follows that $n_5 \equiv 1 \pmod{5}$ and that $n_5 \mid 3 \cdot 23$. This tells us that $n_5 \in \{1, 3, 23\}$. We will show that $n_5 = 1$. Indeed, if $n_5 = 3$, then we get

$$3 \equiv 3 \pmod{5} \not\equiv 1 \pmod{5} \implies n_5 \neq 3.$$

Also, if $n_5 = 23$, then

$$23 \equiv 3 \pmod{5} \not\equiv (\text{mod } 5) \implies n_5 \neq 23.$$

This tells us that $n_5 = 1$ and so P_5 is the unique Sylow 5-subgroup of order 5. Since we already know that P_{23} is the unique Sylow 23-subgroup of order 23, it follows that $P_{23}P_5$ is a subgroup of G and that

$$|P_{23}P_5| = \frac{|P_{23}||P_5|}{|P_{23} \cap P_5|} = |P_{23}||P_5| = 23 \cdot 5$$

since $P_{23} \cap P_5 = 1_G$. Now, we have

$$P_{23}P_5 \cong P_{23} \times P_5 \cong \mathbb{Z}_{23} \times \mathbb{Z}_5 \cong \mathbb{Z}_{23 \cdot 5}$$

since 23 and 5 are relatively prime. Since $\mathbb{Z}_{23 \cdot 5}$ is a cyclic subgroup of G , it follows that $P_{23}P_5$ is also a cyclic subgroup of G of order $23 \cdot 5$ and we are done. ■

- (b) Prove that every Sylow p -subgroup of G is a normal subgroup of G .

Proof. I wasn't sure whether p refers to one of the primes making up the order of G or that we need to prove the result more generally. From part (a), we have $p = 3, 5, 23$ and we have shown that in each instance, we obtain a unique Sylow p -subgroup which is normal in G . ■